YEAR 9 SCIENCE | CHAPTER 1: THE ATOM

What are the particles that make up all matter?

What are the smaller particles that make up atoms?

Draw and label a diagram showing the atomic model.

How do you determine the atomic number of an element?

How can you determine the mass number of an element?

How many different types of atoms exist?

Identify the symbol for each of the following elements.

Element	**Symbol**	**Element**	**Symbol**
Hydrogen		Helium	
Lithium		Beryllium	
Boron		Carbon	
Nitrogen		Oxygen	
Fluorine		Chlorine	
Argon		Potassium	
Calcium		Neon	
Manganese		Iron	
Mercury		Krypton	

List the number of protons, neutrons and electrons are in the following elements

Element	**Protons**	**Neutrons**	**Electrons**
Potassium			
Carbon			
Oxygen			
Sodium			
Fluorine			
Neon			

Draw a diagram representing the following:

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ELEMENT

COMPOUND

MOLECULE

LATTICE

Identify in the following compounds, the elements and their composition

Compound	**Elements**	**Composition**
Water (H_2O)		
Glucose ($C_6H_{12}O_6$)		
Methane (CH_4)		
Table Salt (NaCl)		
Nitrous Oxide (N_2O)		
Calcium Carbonate ($CaCO_3$)		
Potassium Permanganate ($KMnO_4$)		

Horizontal rows in the Periodic table are called:

Vertical columns in the Periodic table are called:

Electrons are configured in an atom in

Complete the table identifying how many electrons can occupy each shell

Shell number	**Electrons it can hold**
1	
2	
3	

Write out the electron configuration of the first 20 elements.

	Group I	Group II	Group III	Group IV	Group V	Group VI	Group VII	Group VIII
Period 1	H							He
Period 2	Li	Be	B	C	N	O	F	Ne
Period 3	Na	Mg	Al	Si	P	S	Cl	Ar
Period 4	K	Ca						

The Period number will always be determined by:

_____

The Group number will always be determined by:

_____

The elements that rarely react are called the:

_____

These are found in Group:

_____

All other atoms want to be like the noble gases, so will _____ _____ electrons.

If an element gains an electron they are termed a _____.

If an element loses an electron they are termed a _____.

Explain using a diagram how sodium (Na) and chlorine (Cl) can rearrange their electrons to become more stable.

Explain using a diagram how ions behave when they are in solution.

Complete the table showing the ionic charges for some common elements.

Element	Atomic number	Number of electrons	Ion formed
H			
Li			
Be			
B			
C			
N			
O			
F			
Na			
Cl			
Mg			
Al			

Work out the chemical formulae for the following compounds

Sodium Fluoride

Sodium Oxide

Sodium Nitride

Magnesium Fluoride

Magnesium Oxide

Magnesium Nitride

Aluminium Fluoride

Aluminium Oxide

Aluminium Nitride

What is meant by a radioisotope?

_____

_____

Why are some nuclei unstable?

What is meant by the half-life of a radioisotope?

How is carbon dating used to estimate the age of some fossils?

Complete the table on types of decay.

Decay	Symbol	Speed	Charge	Penetration	Ionising Ability
Alpha					
Beta					
Gamma					

Give an example of a reaction producing an alpha particle with a diagram.

How can radiation be harmful?

What are some of the uses for radioisotopes?